Fractional-Order Derivative Spectral Transformations Improved Partial Least Squares Regression Estimation of Photosynthetic Capacity From Hyperspectral Reflectance

Guangman Song, Quan Wang, and Jia Jin

***Abstract*— Hyperspectral spectroscopy based on partial least squares regression (PLSR) is an effective tool for monitoring plant photosynthesis. Despite their wide applications, the robustness of PLSR models on tracing photosynthetic capacity, which varies considerably among different species and at different times, has been far less explored, leading to doubt about whether hyperspectral information can accurately predict the capacity across different species and temporal changes. Ordinary applications of PLSR generally make use of original or integer-order derivative transformed reflected spectra, but recent advances in spectral analysis have revealed that fractional-order derivative (FOD) transformed spectra could provide more details of spectral signals. In this study, PLSR models based on FODs coupled with different wavelength selection methods were developed to evaluate whether photosynthetic parameters (Vcmax and Jmax) could be correctly predicted from reflectance spectra. The result indicated that the best PLSR models for the Vcmax and Jmax were obtained based on the sensitive wavelengths selected by stepwise regression using the fractional orders of 1.25 and 1.60, respectively. The optimal PLSR models were able to capture the temporal variabilities of Vcmax and Jmax with the R^2 of 0.62–0.94 and 0.65–0.85, for which the 1605–1845 nm region was consistently used. Meanwhile, these PLSR models have the ability to capture the variations in different species, plant functional types, and biomes. The findings of this study demonstrate that leaf spectra can be successfully used for the timely prediction of variable photosynthetic capacity and provide the fundamentals for monitoring and mapping plant functions from reflected information.**

***Index Terms*— Fractional-order derivative (FOD), hyperspectral reflectance, Jmax, Vcmax, wavelength selection.**

I. Introduction

PHOTOSYNTHETIC capacity, typically represented by the commonly used parameters of the maximum carboxylation rate (Vcmax) and maximum electron transport rate (Jmax) [1], is essential to use the widely applied Farquhar model [2] to understand plant photosynthetic performance at leaf to ecosystem scales under varying environments [3], [4]. Since the two photosynthetic parameters differ extensively among plant functional types and species and vary over different spatial and temporal scales [5], [6], [7], [8], taking their spatiotemporal variations into account is, therefore, critical to comprehending the ecosystem functioning and carbon cycle [9], [10].

Traditional measurements of leaf Vcmax and Jmax using gas exchange systems are cumbersome, time-consuming, and labor-intensive [1], although new approaches are being developed to measure gas exchange rapidly [11], [12], and it remains hard to capture long-term and species-specific variations in them. Alternatively, reflectance spectroscopy, an efficient and nondestructive method, has emerged as a feasible and widely investigated approach to predict leaf Vcmax and Jmax, including crops [13], [14], [15] and tree species [16], [17]. The achieved advances have clearly demonstrated the applicability of reflectance spectra in tracing photosynthetic capacity, as well as plant physiological status.

Partial least squares regression (PLSR), as a standard statistical approach widely used in spectroscopy and chemometric analyses [18], is the most commonly used statistical model for predicting leaf Vcmax and Jmax from spectral information [19], [20], [21], [22] and its feasibility has been proven in a number of previous studies. For instance, Meacham-Hensold et al. [23], Fu et al. [24], and Sexton et al. [25] applied the PLSR to successfully predict the Vcmax and Jmax from reflectance spectra in the leaves of tobacco (*Nicotiana tabacum*). Dechant et al. [26] conducted a PLSR analysis on the estimation of Vcmax and Jmax for a large number of temperate deciduous tree species using reflectance spectra and achieved good accuracy as well.

Despite their successful applications, generating consistent and reliable estimates of photosynthetic capacity from leaf spectroscopy remains challenging. Although PLSR offers opportunities to accurately estimate photosynthetic capacity, the predictive models are typically generated in specific periods and locally, making it challenging to establish robust models across species and over time. For example, Wu et al. [27] found that the PLSR model built for predicting Vcmax on mature leaves in tropical forests needs recalibration

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Guangman Song and Quan Wang are with the Faculty of Agriculture, Shizuoka University, Shizuoka 422-8529, Japan (e-mail: song.guangman@shizuoka.ac.jp; wang.quan@shizuoka.ac.jp).

Jia Jin is with the Institute of Geography and Oceanography, Nanning Normal University, Nanning 530011, China (e-mail: jinjia@nnnu.edu.cn).

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when applied to immature leaves. Additionally, our previous study showed that the PLSR models for estimating leaf Vcmax and Jmax vary annually and seasonally and are not robust at different times [28]. As a result, the robustness of PLSR models across species and time needs a thorough evaluation.

One of the main deficits of current PLSR models for predicting Vcmax and Jmax is the usage of the full range spanning the visible, near-infrared (NIR), and shortwave infrared (SWIR) regions in most previous studies and the importance of each wavelength in models varies [20], [21]. Consequently, the associated physiological signals and underlying physiological mechanisms for reflectance to predict photosynthetic parameters have not yet been detected and understood [23]. Most PLSR models of Vcmax and Jmax are built based on the importance or contributions of the spectral bands according to their loading weights or coefficients generated from full-band regression [25], [29]. In recent progress, statistics-based techniques, such as the least absolute shrinkage and selection operator (lasso), stepwise regression, and genetic algorithm (GA), have been gradually incorporated with the PLSR model to reduce the model dimensionality and improve model performance [30], [31], and for estimating biochemical traits as well [32], [33]. As such, the incorporation of these wavelength selection methods in the PLSR models might be more informative for the predictions of photosynthetic capacity.

What is more, current PLSR models for predicting Vcmax and Jmax rely heavily on the entire domain of the original spectra, leaving a knowledge gap regarding the use of spectral preprocessing techniques as in other spectroscopic models. Various spectral transformations, in particular derivative spectra, are proven to be a powerful mathematical tool that is effective in applying reflectance spectra to estimate a wide variety of plant characteristics [34], [35], [36], due to their ability to reduce background signals and isolate overlapped spectra [37]. Nonetheless, the traditional derivatives, on the other hand, concentrated typically more on the first- and second-order, and even higher orders of integer-order forms, are unable to obtain the gradual tilts contained in the original spectra and thus lead to information omission [38], [39]. Moreover, the first- and second-order transformations that represent the slope and curvature of the spectral curve may result in sharp peaks and severe spectral deformation [40].

By comparison, the fractional-order derivative (FOD) is an extension of conventional integer-order derivatives that enables finer interpolation with arbitrary orders between the integer derivatives of reflectance spectra [41]. Its superiority over integer-order derivatives has previously been reported [40], [42], [43], [44]. This approach has the advantage of effectively retaining the structure of the original curve and providing more details of spectral signals in specified order intervals [45], [46], [47], with a wide range of applications in signal pretreatments [48], [49] and soil parameters [40], [50], [51]. Most recently, the FOD is becoming more attractive in estimating a range of vegetation biochemical traits, with several advances achieved [39], [45], [52]. For instance, Sun et al. [53] used hyperspectral data that was preprocessed with fractional derivatives to estimate the water content of corn leaves and obtained very good results. Jin and Wang [54] developed the FOD-based spectral indices from a composite dataset of various species that can be used to accurately estimate leaf mass per area. These studies have shown considerable promise for characterizing vegetative attributes using reflectance spectra with the FOD pretreatments. Nevertheless, the estimation of physiological traits, such as photosynthetic capacity (Vcmax and Jmax), has not been investigated yet, and thus, the potential of FOD on physiological performance needs further exploration.

In these contexts, the current study provides, to the best of our knowledge, the first attempt to use the PLSR model in combination with FOD preprocessing techniques of reflectance and coupling with different spectral band selection methods to predict the leaf photosynthetic capacity in different species and at different time scales. Accordingly, the objectives of the study are to: 1) investigate how leaf photosynthetic capacity and reflectance vary among species and monthly; 2) develop the applicable PLSR models for Vcmax and Jmax based on the FOD and wavelength selection; and 3) explore the potential of the developed PLSR models for tracking photosynthetic capacity across species and over time. The potential of leaf spectroscopy to track photosynthetic capacity while accounting for species and time differences should lay a solid foundation for extensive surveys of vegetation characteristics using hyperspectral imaging from airborne and space-borne sensors in the future.

II. Materials and Methods

A. Datasets Used in This Study

Six datasets, with synchronous measurements of leaf photosynthetic capacity parameters, i.e., maximum carboxylation rate (Vcmax) and maximum electron transport rate (Jmax), and leaf reflectance, were composited in this study. The detailed descriptions of these datasets are shown in the literature of Burnett et al. [5], Kumagai et al. [15], and Serbin et al. [55], [56]. Briefly, Dataset I contained leaf samples that were collected on the *Glycine max* (GLMA) at the soybean free-air concentration enrichment (SoyFACE) research facility during the growing season on July 10, July 31, and August 23 in 2018, and July 28 and August 25 in 2019 [15]. Dataset II contained the data on both photosynthetic capacity parameters and reflectance of *Guatteria dumetorum*, *Miconia borealis*, *Terminalia amazonia*, and *Vochysia ferruginea* that were collected from the Bosque Protector San Lorenzo (PA-SLZ) NGEE Tropics site in Panama in February 2017 [55]. The deciduous species of *Quercus coccinea* in a mixed pine-oak forest was sampled during the growing season from May 2019 to October 2019 in Dataset III [5]. The leaf samples of 15 managed agricultural and natural plant species in Dataset IV, including cayenne pepper, grape, lemon, date palm, common beet, alfalfa, American pistachio, oat, peach, pomegranate, edible fig, sweet almond, garden tomato, avocado, and Irish potato were collected in June 2013–2015, March 2013, and April 2014 [57]. In addition, Dataset V contained a phenological dataset comprising leaves within growing seasons (generally from May to October) from

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the dominant species of *Fagus crenata* (FACR) in Naeba (138°40′E, 36°51′N). Finally, Dataset VI consists of *Acer nipponicum*, *Acer shirasawanum*, *Acer sieboldianum*, *Betula grossa*, *Carpinus japonica*, *Carpinus tschonoskii*, FACR, *Stewartia monadelpha*, *Stewartia pseudocamellia*, *Styrax japonica*, and *Symplocos coreana,*within growing seasons in Nakakawane (138°06′E, 35°04′N) in Japan.

In summary, for Vcmax, Jmax, and the ratio of Jmax to Vcmax, the composited datasets had 1048, 1121, and 1004 samples, respectively.

B. Fractional-Order Derivative

The FOD, with a wide range of mathematical definitions, such as Riemann–Liouville, Grünwald–Letnikov, and Caputo, has been gradually applied as a positive tool to extract important and sensitive information in a variety of fields [58], [59], [60], [61]. Among these mathematical definitions, the Grünwald–Letnikov algorithm has been widely chosen for FOD calculation since it is simple and comprehensive [38], and this algorithm was used in this study, as well as for spectral transformation. This algorithm uses a gamma function, which is characterized by

$$\Gamma(v) = \int_0^{\infty} \exp(-u)u^{v-1}du = (v-1). \tag{1}$$

The formula of the FOD can be expressed as

$$\begin{aligned}\frac{d^v f(\lambda)}{d\lambda^v} \approx\ & f(\lambda) + (-v)f(\lambda-1) \\ & + \frac{(-v)(-v+1)}{2} f(\lambda-2) + \cdots \\ & + \frac{\Gamma(-v+1)}{m!\Gamma(-v+m+1)} f(\lambda-m)\end{aligned} \tag{2}$$

where v and λ are the fractional order and wavelength, respectively, and $m = [(\lambda/\Delta\lambda]$.

This study investigated FOD spectral transformations with orders ranging from 0.00 to 2.00 with a 0.10 increment and additional orders of 0.25, 0.75, 1.25, and 1.75.

C. Analysis Procedures

We composited the dataset consisting of samples from different species and different months. The analysis of variance (ANOVA) was employed to investigate the effects of species, months, and their interactions on the leaf photosynthetic capacity parameters (Vcmax, Jmax, and Jmax:Vcmax). The variations between species, months, their interactions, and the residuals were quantified using the partitioning of variance through the ratio of the sum of squares of these terms to the total sum of squares in this study. The reflectance variability was represented by the coefficient of variation (CV), which was calculated using the ratio of standard deviation to mean reflectance. To evaluate the effects of months, species, and their interactions on the variations in reflectance, a permutation-based multivariate ANOVA (PERMANOVA) was applied to the matrix of Bray–Curtis dissimilarity among spectral wavelengths. An alpha level of 0.05 was used to test the significance of the nonparametric PERMANOVA performed on the resulting distances (10 000 permutations).

The full dataset was then split into calibration and independent test subsets using a stratified sampling strategy with a ratio of 70%–30%, ensuring that both subsets spanned the full range of observations. The PLSR was developed from the calibration dataset and validated with the independent test subset. The number of components in the PLSR model was determined with Bayesian optimization, which determines the next hyper-parameter value based on the previous results of tested hyper-parameter values and thus avoids unnecessary evaluations [62]. This method is a widely used hyper-parameter optimization method in machine learning models, and the advantage of this method has been demonstrated previously [63], [64]. Ten-fold cross-validation was used for developing the PLSR model to avoid overfitting. Furthermore, a PLSR model uncertainty analysis was conducted to characterize the predictive uncertainty by splitting the original calibration dataset into 70%:30% via 1000 permutations [57], [65].

Wavelength selection approaches are typically employed due to the leaf reflectance data having multidimensionality and being highly dependent on one another [31], [33]. In this study, different wavelength selection methods were employed in the PLSR model, including the lasso [66], GA [67], stepwise regression [68], competitive adaptive reweighted sampling (CARS) [69], and interval random frog (iRF) [70]. The details of wavelength selection approaches have been described by Jin and Wang [33], Liu et al. [71], and Yun et al. [72]. In addition, this study also examined the variable importance in the projection (VIP), for which the variables with a VIP score below the threshold of 1 were excluded [73].

The performance of the PLSR models was quantified using the independent test dataset with the evaluation metrics of the coefficient of determination (R^2), percentage root mean square error (%RMSE), and the ratio of the performance to deviation (RPD). In terms of RPD, the classification criterion from the study of Hong et al. [43] was used in this study, namely, RPD $\geq$ 2.50, 2.00 $\leq$ RPD $<$ 2.50, 1.80 $\leq$ RPD $<$ 2.00, 1.40 $\leq$ RPD $<$ 1.80, 1.00 $\leq$ RPD $<$ 1.40, and RPD $<$ 1.00 represent excellent, very good, good, fair, poor, and very poor predictions, respectively. Generally, a well-performing PLSR model would have a large R^2 and RPD and a low %RMSE.

III. Results

A. Temporal and Species Variations in Leaf Photosynthetic Parameters and Reflectance

Leaf scale Vcmax, Jmax, and the ratio of Jmax to Vcmax (Jmax:Vcmax) varied over months and species ($p < 0.01$), in which the temporal changes were species-specific in Jmax:Vcmax with significant interactions between species and months ($p < 0.05$). The variances partitioning of Vcmax, Jmax, and Jmax:Vcmax are shown in Fig. 1. More variation was explained by differences among species (55.61% and 52.38%) than months (19.29% and 25.54%) for Vcmax and Jmax, respectively. The Jmax:Vcmax, on the other hand,

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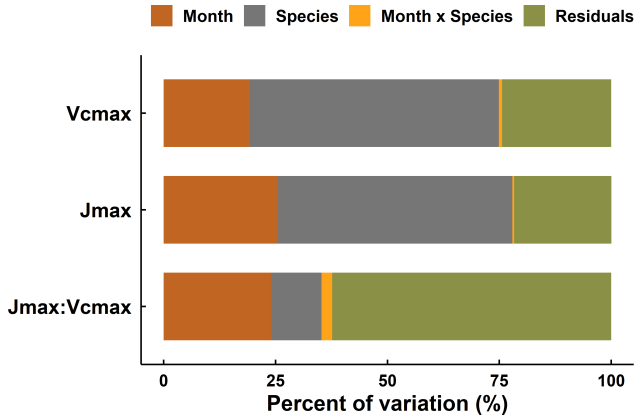


Fig. 1. Variance partitioning of leaf maximum carboxylation rate (Vcmax), maximum electron transport rate (Jmax), and the ratio of Jmax to Vcmax (Jmax:Vcmax) between months, species, their interactions, and residuals for the composite dataset.

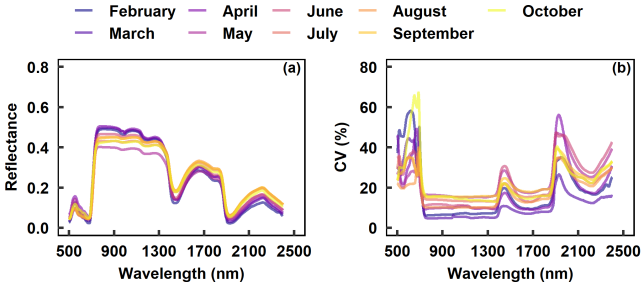


Fig. 2. Mean reflectance (a) and CVs (b) of leaf reflectance in different months for the composite dataset.

showed the opposite trend, with variance partitioning of 11.16% and 24.10% for species and months, respectively. Furthermore, the residuals of Jmax:Vcmax accounted for a large portion of the variations.

Leaf reflectance also varied in different months and species ($p < 0.001$). In detail, the reflectance was lowest in August in the visible (VIS) regions with peaked chlorophyll levels and increased when the leaves flushed in April or yellowed in October (see Fig. 2). In contrast, the reflectance in the NIR regions was low in April and October. In addition, the NIR spectra reached the highest in February and March for tropical species and managed plant species, respectively. The reflectance in the SWIR regions was higher from August to October than in other months. Furthermore, the relative temporal variations in reflectance were found to be greatest in the VIS (CV = 20%–68%) and the SWIR (CV = 10%–60%) but least in the NIR (CV = 5%–30%). In addition, the leaf reflectance varied among different species, and the largest variations also occurred in the VIS and SWIR regions.

B. PLSR Models for Predicting Leaf Photosynthetic Parameters From Reflectance

The PLSR models developed using the full bands or using the specific bands selected by the wavelength selection methods (lasso, GA, stepwise, CARS, iRF, and VIP) from different derivative order spectra are validated by the independent test subset, and the evaluation metrics (i.e., R^2, %RMSE, and RPD) are shown in Fig. 3. For different FOD

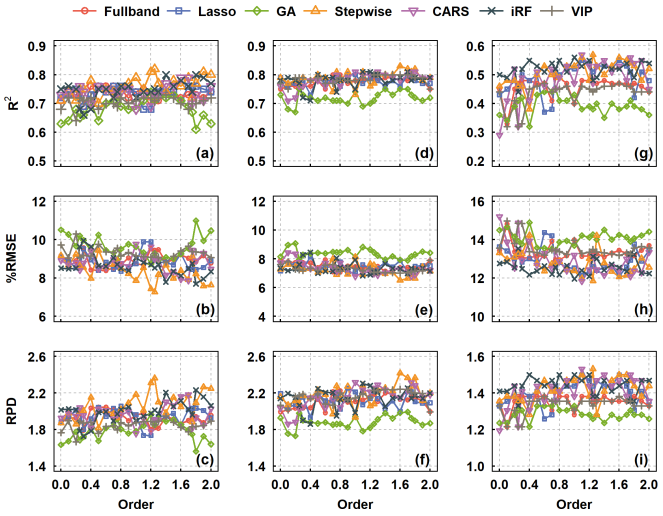


Fig. 3. (a)–(c) Coefficient of determination (R^2), %RMSE over the data range, and the RPD of Vcmax. (d)–(f) Jmax. (g)–(i) Ratio of Jmax to Vcmax using PLSR model coupled with different band selection methods (lasso, GA, stepwise, CARS, iRF, and VIP) in the spectra within different derivative orders from 0.00 to 2.00.

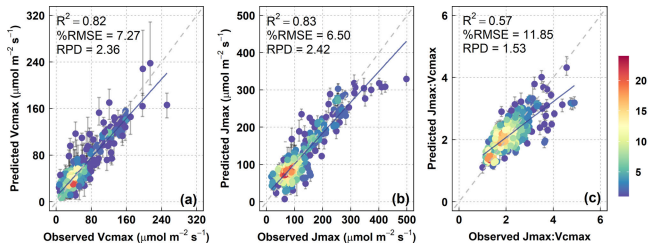


Fig. 4. (a) Performance of the best stepwise PLSR model for the estimation of Vcmax. (b) Jmax. (c) Ratio of Jmax to Vcmax (Jmax:Vcmax, at the optimal derivative orders of 1.25, 1.60, and 1.25, respectively. Error bars represent the 95% confidence intervals.

spectra, the R^2 values of the PLSR models for Vcmax estimation using the full spectral bands varied from 0.69 to 0.75 with the %RMSE of 9.58–8.68 and the RPD of 1.79–2.05. For Jmax, the ranges of R^2, %RMSE, and RPD in the PLSR models without band selection were 0.75–0.80, 7.07–7.88, and 1.98–2.22, respectively. In addition, the R^2 values were from 0.33 (%RMSE: 14.81; RPD: 1.22) to 0.49 (%RMSE: 12.94; RPD: 1.41) for the ratio of Jmax to Vcmax.

On the other hand, when coupling with spectral band selection methods, the stepwise regression performed best, with better performance and fewer spectral bands, for estimating Vcmax, Jmax, and the ratio of Jmax to Vcmax. The stepwise PLSR model had R^2 values ranging from 0.71 to 0.82, the %RMSE from 7.27 to 9.17, and RPD from 1.82 to 2.36 for Vcmax estimation. Similarly, the PLSR model had the R^2 ranges of 0.73 to 0.83 (%RMSE: 6.50–7.60; RPD: 1.93–2.42) at different derivative orders for Jmax. In contrast, this model had a lower performance for the estimation of the ratio of Jmax to Vcmax within different orders, with R^2 of 0.38 to 0.57, %RMSE of 11.85–14.23, and RPD of 1.28–1.53.

Among all the PLSR models developed from different orders of spectra using the stepwise selection strategy, a superiority when using particular derivative orders was identified if compared with the original and integer derivative orders, i.e., first-order and second-order derivatives (see

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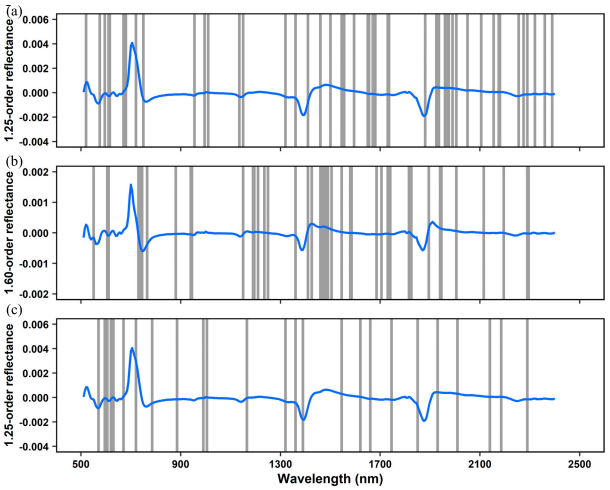


Fig. 5. (a) Informative wavelengths selected by the optimal PLSR model for the estimation of Vcmax. (b) Jmax. (c) Jmax:Vcmax in the composite dataset.

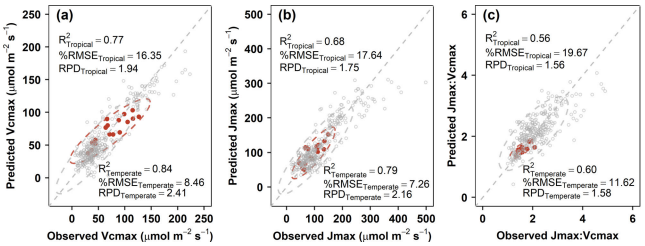


Fig. 6. (a) Performance (R^2, %RMSE, and RPD) of the best PLSR models for Vcmax. (b) Jmax. (c) Jmax:Vcmax estimation in different biomes. Red solid points and gray empty points represent tropical and temperate species, respectively.

Fig. 4). In detail, the derivative order of 1.25 showed the best performance for estimating both Vcmax and the ratio of Jmax to Vcmax, with the R^2 of 0.82 and 0.57, %RMSE of 7.27 and 11.85, and RPD of 2.36 and 1.53, respectively. This model showed consistently good results in the derivative orders from 1.30 to 1.80 for Jmax estimation, with the R^2 and RPD values consistently higher than 0.80 and 2.22 and %RMSE values lower than 7.09, which peaked at the order of 1.60 ($R^2 = 0.83$, %RMSE = 6.50, RPD = 2.42). In addition, 48, 44, and 26 wavelengths were selected by the corresponding optimal PLSR models for the estimation of Vcmax, Jmax, and the ratio, and the wavelengths spread over the entire domain, including VIS, NIR, and SWIR spectral regions, although the majority of wavelengths were distributed within the SWIR spectral domain (see Fig. 5).

C. Performance of the Optimal PLSR Model in Terms of Different Biomes, Months, and Species

This dataset was composed from different datasets covering a variety of biomes, enabling us to validate the robustness of the optimal PLSR model in different biomes (tropical and temperate species) (see Fig. 6). The samples of tropical species in the composite dataset were only collected in February (Dataset II). The R^2 ranges of the optimal PLSR models for Vcmax, Jmax, and the ratio in different biomes were 0.77–0.84, 0.68–0.79, and 0.56–0.60, respectively. The RPD values for both biomes were higher than 1.40. Meanwhile, the

TABLE I

IDENTIFIED WAVELENGTHS OF THE OPTIMAL PLSR MODEL FOR VCMAX, JMAX, AND JMAX:VCMAX IN DIFFERENT BIOMES

Biome	Parameter	Number	Wavelength
Tropical	Vcmax	5	700, 755, 930, 1020, 2045
	Jmax	7	555, 640, 1275, 1640, 1695, 2095, 2215
	Jmax:Vcmax	6	875, 935, 965, 1355, 2355, 2375
Temperate	Vcmax	72	615, 620, 680, 720, 725, 825, 840, 880, 955, 995, 1010, 1015, 1025, 1055, 1115, 1120, 1150, 1155, 1160, 1170, 1195, 1230, 1250, 1275, 1360, 1450, 1455, 1460, 1490, 1500, 1565, 1575, 1590, 1595, 1600, 1605, 1615, 1665, 1690, 1695, 1710, 1730, 1735, 1750, 1780, 1825, 1830, 1845, 1880, 1895, 1915, 1935, 1965, 1995, 2035, 2050, 2070, 2105, 2135, 2140, 2150, 2155, 2165, 2190, 2235, 2275, 2285, 2290, 2320, 2350, 2360, 2375
	Jmax	39	515, 540, 585, 730, 740, 745, 880, 940, 1045, 1190, 1250, 1310, 1410, 1460, 1465, 1480, 1505, 1650, 1655, 1705, 1710, 1725, 1740, 1745, 1775, 1790, 1820, 1885, 1955, 2035, 2110, 2135, 2195, 2255, 2265, 2295, 2315, 2335, 2365, 2370
	Jmax:Vcmax	45	515, 540, 545, 560, 590, 595, 610, 615, 660, 670, 680, 720, 800, 835, 840, 905, 1005, 1015, 1060, 1085, 1145, 1155, 1170, 1215, 1245, 1255, 1290, 1305, 1320, 1410, 1505, 1545, 1605, 1660, 1670, 1780, 1850, 1855, 1955, 1995, 2010, 2115, 2160, 2195, 2380

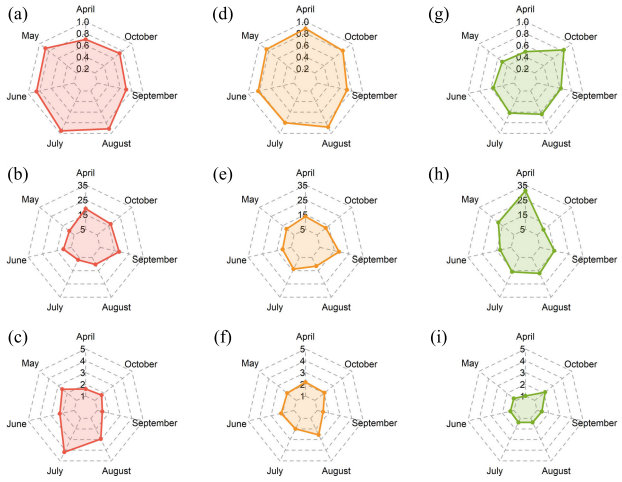


Fig. 7. (a)–(c) Performance (R^2, %RMSE, and RPD) of the best PLSR models in different months for Vcmax. (d)–(f) Jmax. (g)–(i) Jmax:Vcmax.

optimal PLSR models had a relatively poor performance for tropical species compared with temperate species, although many fewer wavelengths were used (see Table I). Overall, the wavelengths chosen for Vcmax estimation in both biomes were concentrated in the red-edge region, 1020 and 2045 nm. In contrast, the wavelengths around 1640 and 1710 nm were consistently selected for Jmax estimation in both biomes.

On the other hand, the species of the temperate biome were sampled in different months, enabling us to further investigate the robustness of the optimal PLSR models in terms of the temporal course, except February and March, when only limited samples were available (see Fig. 7). High accuracies of PLSR models for the Vcmax estimation were obtained consistently from May to August ($R^2 \geq 0.82$, %RMSE $\leq$ 10.44, RPD $\geq$ 2.25), with the highest accuracy in July ($R^2 = 0.94$, %RMSE = 6.83, RPD = 4.18); however,

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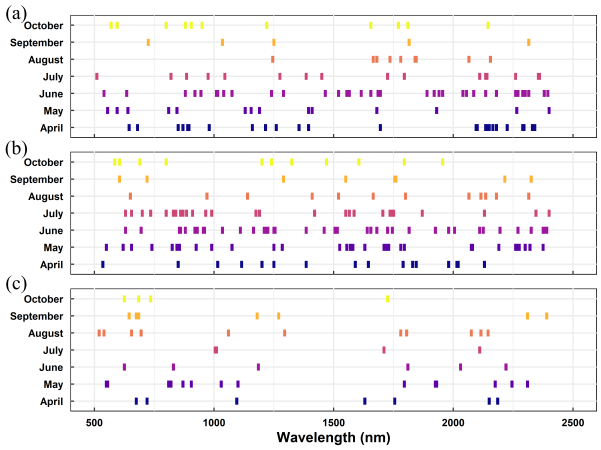


Fig. 8. (a) Distributions of wavelengths selected from the best PLSR model in different months for the estimation of Vcmax. (b) Jmax. (c) Jmax:Vcmax.

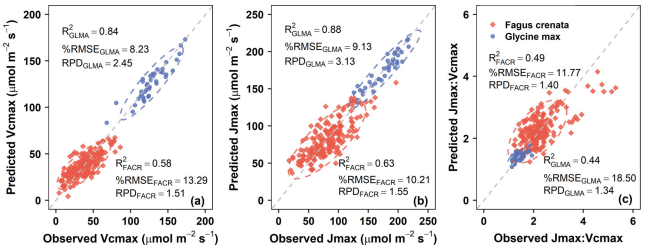


Fig. 9. (a) Performance of the best PLSR models for the Vcmax. (b) Jmax. (c) Jmax:Vcmax estimation in the species of FACR and GLMA.

relatively low accuracies were detected in April (R^2 = 0.62, RPD = 1.59), September (R^2 = 0.63, RPD = 1.44), and October (R^2 = 0.67, RPD = 1.73). As for Jmax, the optimal PLSR models had the R^2 ranging from 0.65 to 0.85, %RMSE from 11.00 to 18.29, and RPD from 1.52 to 2.55, for which the highest and the lowest occurred in August (R^2 = 0.85, RPD = 2.55) and September (R^2 = 0.65, RPD = 1.52), respectively. Relatively low accuracies were, however, found in the estimation of Jmax:Vcmax in different months, except in October, with R^2 of 0.79, %RMSE of 10.57, and RPD of 2.14.

The wavelengths used by the best PLSR models in different months were further explored and are displayed in Fig. 8. In total, 5–36 and 8–39 bands were selected for Vcmax and Jmax estimation, respectively. In detail, for Vcmax estimation, the wavelengths selected for the PLSR models were largely distributed within the NIR and SWIR spectral regions, in which the wavelengths at the 1190–1290 and 1605–1845 nm regions were consistently chosen, irrespective of the months. The selected wavelengths for Jmax estimation, on the other hand, were spread over the entire wavelength domain for each month, although the wavelengths 1605–1845 nm were consistently chosen in different months. In contrast, relatively few wavelengths were selected for the PLSR models when estimating the ratio of Jmax to Vcmax, but the wavelengths varied in different months.

Because of the limited samples, we could only compare the performance of the optimal PLSR models on the species of FACR and GLMA (see Fig. 9). The relatively poor performance of the PLSR model for FACR compared to GLMA

TABLE II

IDENTIFIED WAVELENGTHS OF THE OPTIMAL PLSR MODEL FOR VCMAX, JMAX, AND JMAX:VCMAX IN THE SPECIES OF FACR AND GLMA

Species	Parameter	Number	Wavelength
Fagus crenata	Vcmax	14	595, 600, 645, 770, 810, 835, 885, 900, 1650, 1840, 1885, 1920, 2245, 2280
	Jmax	37	550, 595, 605, 665, 750, 790, 820, 885, 890, 930, 955, 1000, 1085, 1095, 1145, 1150, 1175, 1250, 1515, 1575, 1620, 1665, 1700, 1710, 1775, 1785, 1985, 1995, 2020, 2060, 2130, 2245, 2270, 2300, 2315, 2320, 2385
	Jmax: Vcmax	16	595, 630, 670, 760, 770, 775, 825, 835, 915, 1005, 1155, 1205, 1215, 1275, 1625, 2205
Glycine max	Vcmax	19	520, 570, 590, 605, 705, 930, 980, 1025, 1095, 1205, 1230, 1255, 1270, 1405, 1555, 1620, 2115, 2305, 2325
	Jmax	19	620, 845, 1090, 1100, 1120, 1350, 1360, 1365, 1625, 1640, 1710, 1725, 1740, 1825, 1965, 2065, 2175, 2225, 2315
	Jmax: Vcmax	11	535, 860, 895, 1025, 1075, 1200, 1270, 1920, 2220, 2300, 2325

on estimating Vcmax and Jmax, was observed, considering the R^2, %RMSE, and RPD values. The wavelengths used by these optimal models for the estimation of Vcmax, Jmax, and the ratio are also distributed along the VIS, NIR, and SWIR domains (see Table II). The wavelengths around 600 nm were identified as the commonly important wavelengths for the Vcmax estimation of both species. On the contrary, consistent wavelengths for estimating Jmax were primarily identified in the NIR and SWIR domains at approximately 1090, 1625, 1710, 2065, and 2325 nm.

IV. DISCUSSION

A. FOD for Optimal PLSR Models on Estimating Photosynthetic Parameters

Our results demonstrated that the PLSR model based on FOD spectra and coupled with stepwise regression selection performed best for predicting Vcmax and Jmax from hyperspectral reflectance, with R^2 values of 0.82 and 0.83, respectively. The performance of the PLSR models for Vcmax and Jmax estimation developed in this study was better than the majority of previously published results in terms of R^2 values, with the exception of the studies by Serbin et al. [20], Wu et al. [27], and Ainsworth et al. [74] on Vcmax, and Serbin et al. [20] and Jin et al. [30] on Jmax. Previous studies on using PLSR models to predict leaf photosynthetic parameters from reflectance had reported R^2 for Vcmax and Jmax estimations ranging from 0.39 to 0.89 and from 0.52 to 0.93, respectively [75]. This study was, however, validated based on a much more diverse dataset covering samples from different time scales, and different species, plant functional types, and biomes as well, and the results obtained in this study are, therefore, more reliable in terms of robustness and generalizability.

Furthermore, the results of this study suggest the benefits of using FOD and wavelength selection in the PLSR models for estimating leaf characteristics. FOD with the nonintegers can mine more hidden details, and thus, provides more useful

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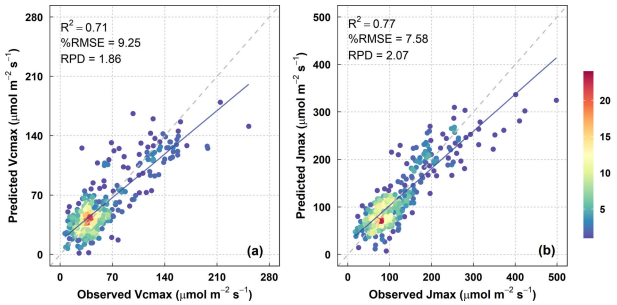


Fig. 10. (a) Measured and predicted optimal PLSR models to estimate Vcmax. (b) Jmax from 1% Gaussian noise spectra.

information from hyperspectral data. Its predictive ability has been proved previously in soil properties and plant biochemicals [40], [42], [45], [50], [51], but unfortunately, not yet for leaf photosynthetic parameters. In this study, the potential of FOD between 0.00 and 2.00 was explored in the PLSR models for the first time to estimate leaf photosynthetic parameters. The promising results demonstrated that the fractional differentiation of spectral data is able to make use of subtle differences to extract informative features, and therefore, enhance the model performance.

The optimal orders of the FOD spectra of the best PLSR models were 1.25 and 1.60 for estimating Vcmax and Jmax, respectively. The results of this study are in agreement with the study by Abulaiti et al. [39], who found that the best accuracy for nitrogen content was observed with optimized spectral indices when the derivative order was beyond 1.50 in cotton leaves. In contrast, Chen et al. [42] and Bhadra et al. [52] demonstrated that the 0.60 and 0.20 order derivatives had the best prediction accuracy when using PLSR models to estimate the nitrogen content of rubber leaves and the chlorophyll of sorghum, respectively. Although no consensus has yet been reached on which optical orders of the FOD spectra should be used, our results, together with previous studies on using FOD to estimate leaf traits, nevertheless highlight that the FOD improves the estimation of leaf traits, although the particular derivative orders may vary and the underlying mechanism needs to be further elucidated.

In addition, high derivative orders are susceptible to noise [76], [77]. In order to validate the stability of the PLSR models developed from FOD, random Gaussian noise with a standard deviation of 1% was artificially added to each leaf reflectance spectrum. The performance of the optimal PLSR models for Vcmax and Jmax was relatively good (see Fig. 10), even though the R^2 values decreased by 13.41% and 7.23%, suggesting the robustness of the optimal PLSR models to trace leaf photosynthetic capacity.

B. Wavelength Selection for PLSR Models on Estimating Photosynthetic Parameters

To the best of our knowledge, very few studies have ever incorporated a variable selection method to pre-select the wavelengths in the PLSR models for the estimation of photosynthetic parameters [20], [30], despite the fact that wavelength selection has become a critical step in the PLSR models [31]. Our results revealed that, among the used wavelength selection methods of this study, the stepwise regression performed best in the PLSR models and obtained better accuracies in estimating photosynthetic parameters than using full spectral bands.

This is in line with previous reports. The stepwise regression selection was found to be the most effective method for identifying informative spectral bands in PLSR models to estimate leaf chlorophyll content in a previous study [33]. Jin et al. [30] have further confirmed that incorporating a stepwise elimination approach in the PLSR models improved the goodness-of-fit to estimate leaf Vcmax and Jmax.

C. Effects of Time on Optimal PLSR Models for Estimating Photosynthetic Parameters

Developing a generally applicable PLSR model to predict leaf photosynthetic capacity parameters from leaf spectroscopy is a primary objective of this study. Although progress has been made in the previous study by Wu et al. [27] conducted in tropical forests for capturing the variation of Vcmax across leaf age, this study provides a more detailed investigation of the influences of time on photosynthetic parameters estimation from reflectance across biomes and plant functional types. In this study, the R^2 values of the PLSR models in estimating Vcmax and Jmax were higher than 0.60, indicating that the PLSR models based on the FOD coupled with stepwise regression wavelength selection have the ability to capture the variations over time. We also performed temporal validations on the optimal PLSR models and confirmed that high validation accuracies were maintained for the estimation of Vcmax and Jmax from June to August (R^2: 0.69–0.84 and 0.67–0.76) but with reduced accuracies in other months ($R^2 < 0.50$), which needs further explicit investigations in future studies.

The sensitive wavelengths selected by the optimal PLSR models for both Vcmax and Jmax estimation were found within the wavelength domain of 1605 and 1845 nm in each month and are possibly associated with the absorption features of cellulose, starch, lignin, and proteins [78]. Such results are close to those of previous studies that declared the importance of reflectance within SWIR domains for Vcmax and Jmax predictions using PLSR models [20], [26], [29]. Furthermore, the important wavelengths for leaf area per mass and nitrogen content are mostly located within the 1605–1845 nm regions [29], which again confirms the importance of these spectral regions in capturing leaf photosynthetic capacity.

D. Are the Optimal PLSR Models Robust in Terms of Different Species, Plant Functional Types, and Biomes for Photosynthetic Parameter Estimation?

The PLSR model performance was species-dependent for Vcmax estimation, with the lowest R^2 and RPD found for FACR species. Such species-specific differences in performance may be attributed to leaf surface properties, but we cannot conclude the generalizability of the optimal PLSR models in terms of species since only two species were examined in this study. The spectral regions between 1605 and 1845 were nevertheless commonly identified for Vcmax and

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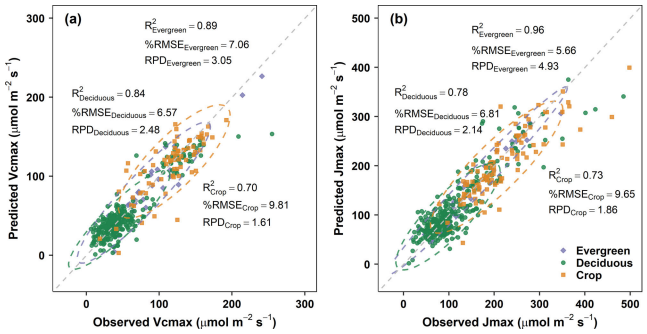


Fig. 11. (a) Performance (R^2, %RMSE, and RPD) of the best PLSR models for Vcmax. (b) Jmax estimation in different plant functional types.

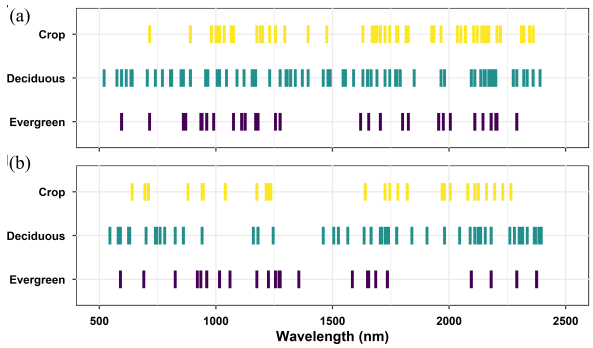


Fig. 12. (a) Identified wavelengths for Vcmax. (b) Jmax estimation of the optimal PLSR models in different plant functional types.

Jmax estimation by the optimal PLSR models in the two examined species. In addition, wavelengths around 600 nm were identified for Vcmax estimation, which is extremely crucial in plant processes [79].

Interestingly, a previous study has explored the stability of PLSR models based on original and integer-order derivatives spectra coupled with different wavelength selection methods for the FACR species but found that the PLSR model is unstable in different growing periods (flushing, maturity, and senescence stages) [28]; however, when we attempted to use the optimal PLSR models of this study for the same species, we obtained a consistently good to excellent performance (R^2 = 0.57–0.88 for Vcmax; R^2 = 0.64–0.91 for Jmax), clearly demonstrating the superiority of incorporating FODs in the PLSR models.

Furthermore, we have examined the applications of the optimal PLSR models on estimating leaf photosynthetic parameters in terms of different plant functional types (see Fig. 11). The Vcmax and Jmax were successfully predicted from spectral information using the optimal PLSR models, capturing the variations in plant functional types, with the R^2 ranges of 0.70–0.89 and 0.73–0.96 for Vcmax and Jmax, respectively. Meanwhile, the PLSR model performance was also dependent on the plant functional types, with evergreen species displaying the highest precision for Vcmax (R^2 = 0.89, %RMSE = 7.06, RPD = 3.05) and Jmax (R^2 = 0.96, %RMSE = 5.66, RPD = 4.93) estimation, accompanied by RPD values that were both higher than 2.50. Despite some uncertainty among plant functional types, the PLSR models were robust in predicting leaf photosynthetic capacity from hyperspectral information in plant functional types. The wavelengths around 710, 960, 1175, 1650, 2150, and 2180 nm were consistently identified for both Vcmax and Jmax estimations in evergreen, deciduous, and crop (see Fig. 12), suggesting the robustness of the optimal PLSR models in plant functional types.

On the other hand, the transferability of optimal PLSR models, rather than just robustness, should also be a consideration with respect to different species and different regions. Previous studies on soil properties have shown that adding a spiking strategy to PLSR calibration can enhance model transferability [80], [81], [82], so it is also worth trying to improve the transferability of optimal PLSR models and thereby provide more accurate estimates of photosynthetic capacity in the context of different species and different geographical scales. In addition, the potential of other machine learning and advanced variable selection methods should be further explored in future studies.

V. Conclusion

This study is the first attempt to apply the PLSR combined with FODs and wavelength selection methods to evaluate the leaf photosynthetic capacity through time from spectroscopy information. The results showed that the PLSR models coupled with FOD transformation and stepwise regression selection increased the performance for estimating photosynthetic parameters, which are resistant to temporal changes as well. The findings suggest that the proposed approach is a powerful tool for evaluating the plant's photosynthetic capacity and has the potential for estimating canopy traits associated with photosynthesis as well, helping with large-scale vegetation growth and productivity monitoring.

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Guangman Song received the Ph.D. degree in environment and energy systems from Shizuoka University, Shizuoka, Japan, in 2022.

She is an Assistant Professor with the Faculty of Agriculture, Shizuoka University. Her research interests include plant eco-physiology and hyperspectral remote sensing.



Quan Wang received the Ph.D. degree in plant ecology from the Institute of Botany, Chinese Academy of Sciences, Beijing, China, in 1997.

He is a Professor and a Faculty Member with the Faculty of Agriculture, Shizuoka University, Shizuoka, Japan. His research interests include remote sensing applications and terrestrial carbon and water cycles.



Jia Jin received the Ph.D. degree in cartography and geographic information system from the Xinjiang Institute of Ecology and Geography, Chinese Academy of Sciences, Ürümqi, China, in 2017.

He is an Assistant Professor with Nanning Normal University, Nanning, Guangxi, China. His research interests include radiative transfer model and remote sensing.

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